

LaTeX Complete Reference & Cheatsheet

Structure • Fonts & Text • Mathematics • Tables • Figures • Lists • Bibliography • TikZ • Custom Commands • Advanced

July 15, 2026 — BUET Exam Resource

DOCUMENT STRUCTURE

Document Classes

```
\documentclass[options]{class}

% -- Classes -----
% article    papers, reports, short docs
% report     longer docs, theses
% book       books (has \chapter)
% beamer     presentations / slides
% letter     formal letters
% standalone single figure / diagram

% -- Common Options -----
% Font size: 10pt 11pt 12pt (default: 10pt)
% Paper size: a4paper letterpaper a5paper
% Columns:   onecolumn twocolumn
% Sides:     oneseide twoside
% Misc:      draft final titlepage
```

Minimal Document Skeleton

```
\documentclass[12pt,a4paper]{article}
%%-- PREAMBLE -----
\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{amsmath,amssymb}

\title{Document Title}
\author{Name \and Co-Author}
\date{\today}           % or \date{} for no date

%%-- BODY -----
\begin{document}

\maketitle
\begin{abstract}
  Abstract text here.
\end{abstract}
\tableofcontents
\newpage

\section{Introduction}
Hello, \LaTeX{}!

\end{document}
```

Compile Commands

pdflatex file.tex	% compile -> PDF
pdflatex file.tex	% again for TOC/refs
xelatex file.tex	% Unicode + system fonts
lualatex file.tex	% Lua scripting
latexmk -pdf file.tex	% auto multi-pass
bibtex file	% BibTeX references
biber file	% biblatex references

ESSENTIAL PACKAGES

Package	Purpose
amsmath	Math environments (align, equation...)
amssymb	Symbols: R, N, V, E
amsthm	theorem, proof environments
mathtools	Enhanced math (dcases, prescript...)
bm	Bold math: \bm{x}
cancel	Math strikethrough/cancel
geometry	Page margins and layout
graphicx	\includegraphics
xcolor	Colors everywhere
hyperref	Clickable links in PDF
fancyhdr	Custom headers & footers
titlesec	Customize section headings
enumitem	List customization
booktabs	Professional table rules
multirow	Multi-row table cells
tabularx	Auto-width table columns
longtable	Tables spanning multiple pages
colortbl	Colored table rows/cells
array	Enhanced column specifiers
listings	Source code with highlighting
tcolorbox	Fancy colored boxes
mdframed	Framed environments
caption	Custom figure/table captions
subcaption	Sub-figures and sub-tables
float	[H] exact placement specifier
wrapfig	Text-wrapped figures
tikz	Drawings, diagrams, flowcharts
pgfplots	Data plots and charts
biblatex	Modern bibliography management
cleveref	Smart cross-refs: \cref{key}
microtype	Micro-typography improvements
setspace	Line spacing control
multicol	Multi-column layout
soul	Highlight, strikethrough
ulem	Underline, strikethrough text
siunitx	SI units typesetting
xparse	Advanced command definitions
ifthen	Conditional compilation
fontenc	Output font encoding [T1]
inputenc	Input encoding [utf8]
lmodern	Latin Modern fonts (better PDF)

TEXT FORMATTING

Font Styles

```
\textbf{bold}           \textit{italic}
\texttt{typewriter}     \textsf{sans-serif}
\textrm{roman/serif}    \textsc{Small Caps}
\textsl{slanted}        \textnormal{normal}
\underline{underline}   \emph{emphasized}
% \emph toggles: in normal text -> italic;
% inside italic -> upright

% Scope-based (use in groups):
{\bfseries bold}        {\itshape italic}
{\ttfamily mono}       {\sffamily sans}
```

```
% Combine styles:
\textbf{\textit{bold italic}}
```

```
% Strikethrough:
\usepackage{ulem}       \sout{strikethrough}
\usepackage{soul}       \st{strikethrough}
```

Font Sizes (smallest -> largest)

```
{\tiny      tiny text    } % ~5pt
{\scriptsize scriptsize } % ~7pt
{\footnotesize footnotesize } % ~8pt
{\small     small       } % ~9pt
{\normalsize normalsize  } % 10pt (default)
{\large     large        } % ~12pt
{\Large     Large        } % ~14pt
{\LARGE     LARGE        } % ~17pt
{\huge     huge          } % ~20pt
{\Huge      Huge         } % ~25pt

% Custom size:
\fontsize{14pt}{18pt}\selectfont % {size}{baseline}
```

— 1 —

Text Alignment

```
\begin{center} centered text \end{center}
\begin{flushleft} left-aligned \end{flushleft}
\begin{flushright} right-aligned \end{flushright}

% Declarations (inside groups/envs):
\centering % center
\raggedright % left-align (ragged right)
\raggedleft % right-align
% Default: fully justified
```

Colors (xcolor package)

```
\usepackage{xcolor}

% Named colors (predefined):
% red, green, blue, cyan, magenta, yellow,
% black, white, gray, darkgray, lightgray,
% brown, lime, olive, orange, pink, purple,
% teal, violet

\textcolor{red}{colored text}
\textcolor{blue!50}{50% blue + 50% white}
\textcolor{red!40!blue}{40% red + 60% blue}
{\color{green} green scope}
\colorbox{yellow}{yellow background}
\colorbox{red}{yellow}{framed + colored bg}
\pagecolor{white} % page background color

% Define custom colors:
\definecolor{myblue}{RGB}{30,100,180}
\definecolor{mygr}{HTML}{1A8040}
\definecolor{lg}{gray}{0.8} % 0=black, 1=white
\definecolor{mc}{cmyk}{0,0.5,1,0}
% Mix: red!30!blue = 30% red + 70% blue
```

Spacing

```
%-- Horizontal -----
\hspace{1.5cm} \hspace*{1cm} % * = forced
% non-breaking space
\quad % 1em horizontal space
\qquad % 2em horizontal space
\hfill % fill available space
\hrulefill % fill with rule
\dotfill % fill with dots

%-- Vertical -----
\vspace{1cm} \vspace*{1cm} % * = forced
\smallskip % ~3pt
\medskip % ~6pt
\bigskip % ~12pt
\vfill % fill vertical space
\\ % line break
\\[6pt] % line break + extra 6pt
\newline
\par % start new paragraph
\newpage % force next page
\clearpage % flush floats, then new page

%-- Line Spacing -----
\usepackage{setspace}
\singlespacing \onehalfspacing \doublespacing
\setstretch{1.3} % custom factor
\begin{spacing}{1.5}...\end{spacing}
```

Special Characters

```
%-- Must escape (LaTeX reserved) -----
\$ \% \& \# \_ \{ \}
\~{} \^{}
\textbackslash % backslash \
\textless % less-than <
\textgreater % greater-than >
\textasciitilde % tilde ~
\textasciicircum % caret ^

%-- Dashes -----
- % hyphen: high-level
-- % en-dash: pages 5--10
--- % em-dash: an idea---like this
$-$ % minus sign (in math)

%-- Quotes -----
'single' 'double'
\lq single\rq \lq\lq double\rq\rq

%-- Ellipsis -----
\ldots % ... (text: dots at baseline)
\cdots % ... (math: centered dots)
\vdots % : (math: vertical dots)
\ddots % . (math: diagonal dots)
\dots % context-sensitive (amsmath)

%-- Symbols -----
\textregistered % (R)
\textcopyright % (C)
\texttrademark % (TM)
\S \P % Sec Par
\dag \ddag % + ++
\pounds \euro \yen % GBP EUR JPY
```

DOCUMENT SECTIONING

```
% book and report only:
\part{Part I} % level -1
\chapter{Chapter Title} % level 0

% All classes:
\section{Section} % level 1
\subsection{Subsection} % level 2
\subsubsection{Sub-sub} % level 3
\paragraph{Paragraph} % level 4 (inline, no #)
\subparagraph{Sub-para} % level 5 (inline, no #)

% Starred = unnumbered + excluded from TOC:
\section*{Unnumbered Section}
\subsection*{Unnumbered Subsection}

% Add manually to TOC:
\addcontentsline{toc}{section}{Custom Entry}

% Generate TOC/LOF/LOT:
\tableofcontents
\listoffigures
\listoftables

% Appendix (resets counters: A, B, C, ...):
\appendix
\section{Proof Details} % -> "A. Proof Details"

% Page numbering:
\pagenumbering{arabic} % 1, 2, 3 (default)
\pagenumbering{roman} % i, ii, iii
\pagenumbering{Roman} % I, II, III
\pagenumbering{alph} % a, b, c
\setcounter{page}{1} % reset page counter
```

PAGE LAYOUT

Geometry Package

```
\usepackage[
  a4paper,           % paper: a4, letter, a5
  top=2.5cm, bottom=2.5cm,
  left=3cm, right=2cm,
  % OR: margin=2cm (all sides equal)
  hmargin={3cm,2cm}, % {left, right}
  vmargin={2cm,3cm}, % {top, bottom}
  headheight=15pt,
  includeheadfoot,   % body includes head/foot
  landscape,          % rotate paper
]{geometry}
\newgeometry{margin=1cm} % change mid-document
\restoregeometry        % restore original
```

Headers & Footers (fancyhdr)

```
\usepackage{fancyhdr}
\pagestyle{fancy}
\fancyhf{} % clear all fields

% Positions: L=left C=center R=right
\fancyhead[L]{Left Header}
\fancyhead[C]{\LaTeX\ Reference}
\fancyhead[R]{\thepage}
\fancyfoot[C]{--- \thepage ---}
% For two-sided: O=odd, E=even pages
\fancyhead[RO,LE]{\thepage}

\renewcommand{\headrulewidth}{0.4pt} % horiz rule
\renewcommand{\footrulewidth}{0pt}   % no footer rule

% Built-in page styles:
\pagestyle{plain} % page number only
\pagestyle{empty} % no header or footer
\pagestyle{headings} % auto chapter/section
\thispagestyle{empty} % this page only

% Useful tokens in headers:
\thepage % current page number
\leftmark % chapter/section (book)
\rightmark % section/subsection
\today % current date
```

Multi-Column Layout

```
\usepackage{multicol}
\setlength{\columnsep}{1cm} % gap between cols
\setlength{\columnseprule}{0.4pt} % vertical rule

\begin{multicols}{2} % balanced columns
  Content...
\end{multicols}
\begin{multicols*}{3} % unbalanced (fills left first)
  Content...
\end{multicols*}
\columnbreak % force column change

% Built-in two-column (less flexible):
\documentclass[twocolumn]{article}
\twocolumn \onecolumn % switch mid-document
```

MATHEMATICS

Math Modes

```
%-- Inline -----
$f(x) = x^2 + 1$ \ (f(x) = x^2 + 1\ )

%-- Display, numbered -----
\begin{equation}
  E = mc^2 \label{eq:einstein}
\end{equation}

%-- Display, no number -----
\begin{equation*}
  \int_0^\infty e^{-x} dx = 1
\end{equation*}
\ [ f(x) = \int_0^x g(t) dt \ ] % shorthand

%-- Multi-line, aligned (B = alignment pt) -
\begin{align}
  a &= b + c \\
  x &= y - z \label{eq:line2}
\end{align}
\begin{align*} % no equation numbers
  \sin^2\theta + \cos^2\theta &= 1 \\
  e^{i\pi} + 1 &= 0
\end{align*}

%-- Gather (centered rows, no alignment) --
\begin{gather}
  f(x) = ax^2 + bx + c \\
  g(x) = \sqrt{x}
\end{gather}

%-- Multiline (one long equation, split) --
\begin{multline}
  f = a + b + c + d \\
  + e + f + g
\end{multline}

%-- Subequations (1a), (1b), ... -----
\begin{subequations}
  \begin{align}
    a &= 1 \\
    b &= 2
  \end{align}
\end{subequations}

%-- Number control -----
\tag{1.5} % custom number label
\notag % suppress number on this line
\nonumber % same as \notag
\numberwithin{equation}{section} % 1.1, 1.2...
```

Greek Letters

```
%-- Lowercase -----
\alpha \beta \gamma \delta
\epsilon \varepsilon \zeta \eta
\theta \vartheta \iota \kappa
\lambda \mu \nu \xi
\pi \varpi \rho \varrho
\sigma \varsigma \tau \upsilon
\phi \varphi \chi \psi \omega

%-- Uppercase (only non-Latin letters exist)
\Gamma \Delta \Theta \Lambda
\Xi \Pi \Sigma \Upsilon
\Phi \Psi \Omega

% Note: A, B, E, Z, H, I, K, M, N, O, P, T, X
% are regular Latin in math mode.
```

Fractions, Powers, Roots

```
\frac{a}{b} % a/b (adapts to context)
\dfrac{a}{b} % display-style fraction
\textfrac{a}{b} % text-style fraction
\cfrac{1}{1+\cfrac{1}{x}} % continued fraction

x^{2n} x^2 % superscript
x_{i,j} x_i % subscript
x^{a}_{b} % both (order doesn't matter)
\prescript{n}{k}{A} % left super + sub
\sqrt{x} % square root
\sqrt[n]{x} % n-th root
x^{1/2} % inline fraction form
x^{x^x} % nested powers
```

Sums, Products, Integrals

```
\sum_{i=1}^n x_i      % Sigma summation
\prod_{k=1}^n k!      % Pi product
\coprod              % coprod coproduct

\int_a^b f(x)\,dx     % int integral
\int_{-\infty}^{+\infty}
\iint \iiint \iiint % iint iiint (multi-dim)
\oint_C \vec{F}\cdot d\vec{r} % oint contour

% Force display-style limits inline:
$\displaystyle\sum_{i=1}^n x_i$
$\sum\limits_{i=1}^n x_i$ % above/below forced
$\sum\nolimits_{i=1}^n x_i$ % side forced
```

Limits & Standard Functions

```
\lim_{x \to 0} f(x)      \lim_{n \to \infty}
\limsup_{n \to \infty} a_n \liminf_{n \to \infty} a_n

% Trigonometric:
\sin x \cos x \tan x \cot x \sec x \csc x
\arcsin \arccos \arctan
\sinh \cosh \tanh

% Exponential / Logarithmic:
\exp(x) e^x
\log x \log_2 x \ln x \lg x

% Other:
\max_k \min_{x \in S} \sup \inf
\det(A) \dim V \ker T \hom(A,B)
\gcd(a,b) \deg f \Pr[E] \arg z

% Define custom operator:
\DeclareMathOperator{\sgn}{sgn}
\DeclareMathOperator{\tr}{tr}
\DeclareMathOperator*{\argmin}{arg\,min} % * -> limits
\DeclareMathOperator*{\argmax}{arg\,max}
```

Binary Operators & Relations

```
%-- Arithmetic -----
+ - \times \div \cdot \pm \mp
\oplus \ominus \otimes \oslash \odot
\cup \cap \uplus \sqcup \sqcap
\vee \wedge \setminus \wr \circ
\bullet \star \ast \bigtriangleup

%-- Comparison -----
= \neq < > \leq \geq \leq \geq
\ll \gg \lll \ggg
\equiv \not\equiv \sim \simeq
\approx \cong \propto \asymp \doteq

%-- Set / Logic -----
\in \notin \ni \subset \supset
\subseteq \supseteq \subsetneq
\sqsubset \sqsubseteq \sqsupseteq
\parallel \perp \mid \nmid
\vdash \dashv \models \Vdash

%-- Using \not for negation -----
\not= \not\in \not\subset \not<
```

Arrows

```
%-- Single -----
\to \gets % -> <-
\rightarrow \leftarrow % same
\leftrightarrow % <->
\longrightarrow % --->
\longleftarrow % <---
\longleftarrow \rightarrow % <--->
\uparrow \downarrow \updownarrow
\nearrow \searrow \swarrow \nwarrow
\mapsto \longmapsto % |-> x |-> f(x)
\hookleftarrow \hookrightarrow
\rightharpoonup \leftharpoonup
\rightharpoons % <-> equilibrium

%-- Double -----
\Rightarrow \Leftarrow \Leftrightarrow
\Longrightarrow \Longleftarrow
\Longleftrightarrow
\Uparrow \Downarrow \Updownarrow
\implies \iff % => <=> (amsmath)

%-- Labelled arrows (amsmath) -----
\xrightarrow[below]{above}
\xleftarrow[below]{above}

%-- Negated (amssymb) -----
\nrightarrow \nleftarrow \nleftrightarrow
```

Math Accents & Decorations

```
%-- Accents -----
\hat{x} \widehat{xyz} % x xyz
\tilde{x} \widetilde{xyz} % x xyz
\bar{x} \overline{xyz} % x x+y
\vec{x} \overrightarrow{AB} % x
\dot{x} \ddot{x} \dotted{x} % x x
\acute{x} \grave{x} \check{x} \breve{x}
\mathring{A} % Angstrom

%-- Overbrace / Underbrace -----
\overbrace{a+b+c}^{n \text{ terms}}
\underbrace{x_1+\cdots+x_n}_{n \text{ terms}}

%-- Bars & Lines -----
\overline{a+b} \underline{c+d}

%-- Stacking -----
\overset{?}{=}{=} % ? above =
\underbrace{\text{def}}{=} % text below =
\stackrel{!}{=} % old-style (avoid)

%-- Cancel (cancel package) -----
\cancel{x} % x with / through it
\bcancel{x} % x with \ through it
\xcancel{x} % x with x through it
\cancelto{0}{x} % x with arrow pointing to 0
```

Delimiters

```
%-- Auto-sizing: wrap with \left...\right --
\left( \right) % ( ) parentheses
\left[ \right] % [ ] brackets
\left\{ \right\} % { } braces
\left| \right| % |x| absolute value
\left\| \right\| % ||x|| norm
\left\lfloor \right\rfloor % floor ceiling
\left\lceil \right\rceil % ceiling ceiling
\left\langle \right\rangle % <> angle bracket
\left. \right|_a^b % one-sided (.)

%-- Fixed sizes (smallest -> largest) ----
\bigl( \bigr) \Bigl( \Bigr)
\biggl( \biggr) \Biggl( \Biggr)
% Example: \bigl[ x + y \bigr]^2

%-- Declare paired (mathtools) -----
\DeclarePairedDelimiter{\abs}{\lvert}{\rvert}
\abs{x} % |x| (fixed)
\abs*x % |x| (auto-sized)
```

Math Fonts

```

\mathbf{ABCxyz}      % Bold (upright)
\bm{ABCxyz}          % Bold (italic, bm pkg)
\mathit{ABCxyz}      % Italic (default in math)
\mathrm{ABCxyz}      % Upright roman
\mathsf{ABCxyz}      % Sans-serif
\mathtt{ABCxyz}      % Typewriter
\mathcal{ABCXYZ}      % Calligraphic (UPPER only)
\mathbb{NZQRCF}      % Blackboard bold (amssymb)
\mathfrak{ABCabc}    % Fraktur (amssymb)
\text{normal text}   % text inside math (amsmath)
\intertext{paragraph} between align rows
    
```

Matrices (amsmath)

```

\begin{matrix} a & b & \backslash c & d \end{matrix} % bare
\begin{pmatrix} a & b & \backslash c & d \end{pmatrix} % ( )
\begin{bmatrix} a & b & \backslash c & d \end{bmatrix} % [ ]
\begin{Bmatrix} a & b & \backslash c & d \end{Bmatrix} % { }
\begin{vmatrix} a & b & \backslash c & d \end{vmatrix} % | |
\begin{Vmatrix} a & b & \backslash c & d \end{Vmatrix} % || ||
\begin{smallmatrix} a&b\backslash c&d \end{smallmatrix} % inline

% General (use any delimiter):
\left[\begin{array}{cc}
a & b \backslash c & d \\
\end{array}\right]

% Dots:
\begin{pmatrix}
a_{11} & \cdots & a_{1n} \\
\vdots & \ddots & \vdots \\
a_{m1} & \cdots & a_{mn}
\end{pmatrix}

% Augmented (with vertical bar):
\left[\begin{array}{cc|c}
1 & 0 & 3 \\
0 & 1 & 5
\end{array}\right]
    
```

Cases & Piecewise Functions

```

f(x) = \begin{cases}
x^2 & \text{if } x > 0 \\
0 & \text{if } x = 0 \\
-x^2 & \text{if } x < 0
\end{cases}

% dcases (mathtools): display-size conditions
g(n) = \begin{dcases}
\frac{n}{2} & n \equiv 0 \pmod{2} \\
3n + 1 & n \equiv 1 \pmod{2}
\end{dcases}

% rcases: brace on right side
\begin{rcases}
a + b \backslash c + d \\
\end{rcases} = 0
    
```

Spacing in Math Mode

```

a\!b      % negative thin (-3mu)
a\,b      % thin space (3mu) <- use before dx
a\:b      % medium space (4mu)
a\;b      % thick space (5mu)
a\ b      % normal interword space
a\quad b  % 1em (word gap)
a\qquad b % 2em

% Phantom (invisible space):
\phantom{xyz} % exact size of "xyz"
\hphantom{xyz} % horizontal size only
\vphantom{X} % vertical size only (align)

% Best practice:
f(x)\,dx % thin space before dx (correct)
f\colon X\to Y % colon in function def
    
```

Theorem Environments (amsthm)

```

\usepackage{amsthm}

% Three styles:
\theoremstyle{plain} % bold head, italic body
\newtheorem{theorem}{Theorem}
\newtheorem{lemma}[theorem]{Lemma} % shared counter
\newtheorem{corollary}[corollary]{Corollary} % per sect.
\newtheorem{proposition}[proposition]{Proposition}

\theoremstyle{definition} % bold head, normal body
\newtheorem{definition}{Definition}
\newtheorem{example}{Example}

\theoremstyle{remark} % italic head, normal body
\newtheorem*{remark}{Remark} % * = unnumbered

% Usage:
\begin{theorem}[Optional Label]
  For all  $x \in \mathbb{R}$ ,  $x^2 \geq 0$ .
\end{theorem}

\begin{proof}
  Since  $x^2 = x \cdot x$  and squares are
  non-negative, the result follows.
\end{proof}

% Proof automatically adds QED symbol #
% Custom proof label:
\begin{proof}[Proof of Theorem~\ref{thm:main}]
  ...
\end{proof}
    
```

Common Math Symbols

```

%-- Sets / Logic -----
\mathbb{N} \mathbb{Z} \mathbb{Q}
\mathbb{R} \mathbb{C} \mathbb{F}
\emptyset \varnothing % {}
\forall \exists \nexists % forall exists not exists
\in \notin \ni
\subset \supset \subseteq \supseteq
\cup \cap \setminus % u n \

%-- Logic -----
\neg \lnot \land \wedge % not /\
\lor \vee % \/
\top \bot % T _
\therefore \because % .. because

%-- Calculus -----
\infty \partial \nabla % inf d del
\triangle \Box % tri sq

%-- Misc -----
\hbar \ell % hbar l
\Re \Im % Re Im (real, imaginary)
\wp % wp (Weierstrass p)
\aleph \beth % aleph beth (cardinalities)
\prime % ' (use as f'prime or f')
\angle \measuredangle % angle mangle
\perp \parallel % _ / |
\infty \propto % inf propto
    
```

LISTS

```
%-- Bullet List -----
\begin{itemize}
  \item First item
  \item Second item
  \item[--] Custom dash label
  \item[{$\star$}] Star bullet
\end{itemize}

%-- Numbered List -----
\begin{enumerate}
  \item First % 1.
  \item Second % 2.
  \begin{enumerate}
    \item Nested a % (a)
    \item Nested b
  \end{enumerate}
\end{enumerate}

%-- Description List -----
\begin{description}
  \item[TCP] Transmission Control Protocol
  \item[UDP] User Datagram Protocol
\end{description}

%-- enumitem Customization -----
\usepackage{enumitem}

\begin{enumerate}[label=\alph*.] % a. b. c.
\begin{enumerate}[label=(\roman*)] % (i) (ii) (iii)
\begin{enumerate}[label=\Roman*.] % I. II. III.
\begin{enumerate}[label=\Alph*.] % A) B) C)
\begin{enumerate}[label=\arabic*.] % 1) 2) 3)

% Spacing control:
\begin{itemize}[noitemsep] % remove extra space
\begin{itemize}[nosep] % remove all spacing
\begin{itemize}[topsep=0pt, itemsep=2pt, parsep=0pt]
\begin{itemize}[leftmargin=2cm] % custom indent

% Resume previous counter:
\begin{enumerate}[resume]
% Start at custom number:
\begin{enumerate}[start=5]
```

Professional Tables (booktabs)

```
\usepackage{booktabs}
\begin{tabular}{@{}lcr@{}} % @{} removes edge space
  \toprule
  Name & Age & Score \\
  \midrule
  Alice & 20 & 95 \\
  Bob & 22 & 87 \\
  \midrule
  Mean & & 91 \\
  \bottomrule
\end{tabular}

% Rules:
\toprule % thick rule at top
\midrule % thin rule in middle
\bottomrule % thick rule at bottom
\cmidrule{2-3} % partial rule cols 2-3
\cmidrule(lr){2-3} % trimmed on left & right
\addlinespace[3pt] % extra vertical space
```

Spanning Cells

```
\usepackage{multirow}

% Horizontal span:
\multicolumn{n}{alignment}{content}
% n = columns to span, alignment = l, c, r

% Vertical span:
\multirow{n}{width}{content}
% n = rows to span, width = * (auto)

% Example:
\begin{tabular}{|c|c|c|}
  \hline
  \multicolumn{2}{|c|}{Merged A+B} & C \\
  \hline
  \multirow{2}{*}{Tall} & B & C \\
  & E & F \\
  \hline
\end{tabular}
```

TABLES

Tabular Basics

```
\begin{tabular}[pos]{col-spec}
  ...rows...
\end{tabular}
% pos: t (top), b (bottom), c (center)

%-- Column Specifiers -----
% l left-aligned
% c center-aligned
% r right-aligned
% | single vertical rule
% || double vertical rule
% p{width} paragraph, top-aligned
% m{width} paragraph, middle (array pkg)
% b{width} paragraph, bottom (array pkg)
% @{expr} replace inter-column space
% @{} remove inter-column space
% *{n}{spec} repeat spec n times
% Example: {|l|*{3}{c}|r|} = |l|ccc|r|

%-- Row & Cell Syntax -----
% @ separates columns
% \\ ends a row
% \hline draws a full horizontal rule
```

Table Float, tabularx, colortbl

```
%-- Table float -----
\begin{table}[htbp]
  \centering
  \caption{Caption goes here.}
  \label{tab:results}
  \begin{tabular}{...}
    ...
  \end{tabular}
\end{table}
% Float specifiers: h=here, t=top, b=bottom,
% p=page, !=override, H=EXACTLY here (float pkg)
Table~\ref{tab:results} % -> "Table 1"

%-- Auto-width column X -----
\usepackage{tabularx}
\begin{tabularx}{\linewidth}{lXr}
  Fixed & Auto-expanding & Fixed \\
\end{tabularx}

%-- Multi-page table -----
\usepackage{longtable}
\begin{longtable}{lc}
  \caption{Long Table} \label{tab:long} \\
  \toprule A & B \\ \midrule
  \endfirsthead % end first-page header
  \midrule A & B \\ \midrule
  \endhead % end continuation header
  \bottomrule \endfoot % footer every page
  ...rows...
\end{longtable}

%-- Colored cells/rows -----
\usepackage{colortbl}
\rowcolor{gray!20} % whole row (in row)
\cellcolor{yellow!50}Cell % single cell
% In column spec: >{\columncolor{blue!10}}c
\arrayrulecolor{blue} % color the rules
```

FIGURES & IMAGES

Including Graphics

```
\usepackage{graphicx}

\includegraphics{filename}           % no extension
\includegraphics[width=0.8\linewidth]{fig.pdf}

%-- Options (comma-separated in []) -----
% width=5cm / width=0.8\textwidth
% height=4cm
% scale=0.5           (scale factor)
% angle=90            (rotate degrees CCW)
% trim=L B R T        (trim: left bottom right top)
% clip                (clip after trim)
% keepaspectratio      (maintain aspect ratio)
% draft               (show box only)
% page=2              (for multi-page PDF)

%-- Useful widths -----
\textwidth   \linewidth   \columnwidth
0.5\textwidth \paperwidth

%-- Set graphics path -----
\graphicspath{{./figures/}{./images/}}
```

Figure Float

```
\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.8\linewidth]{img}
  \caption{Figure description here.}
  \label{fig:myfig}
\end{figure}

See Figure~\ref{fig:myfig}. % -> "Figure 1"

% Force exact placement:
\usepackage{float}
\begin{figure}[H] % H = HERE exactly
```

Subfigures

```
\usepackage{caption}
\usepackage{subcaption}

\begin{figure}[htbp]
  \centering
  \begin{subfigure}[b]{0.45\linewidth}
    \includegraphics[width=\linewidth]{a}
    \caption{First image} \label{fig:a}
  \end{subfigure}
  \hfill
  \begin{subfigure}[b]{0.45\linewidth}
    \includegraphics[width=\linewidth]{b}
    \caption{Second image} \label{fig:b}
  \end{subfigure}
  \caption{Overall caption.} \label{fig:both}
\end{figure}

% Reference: Figure~\ref{fig:a}
```

Wrap Figures & Caption Config

```
\usepackage{wrapfig}
\begin{wrapfigure}{r}{0.4\linewidth} % r=right, l=left
  \centering
  \includegraphics[width=\linewidth]{img}
  \caption{Wrapped figure.}
\end{wrapfigure}
Text flows around the figure here...

%-- Caption customization -----
\usepackage{caption}
\captionsetup{
  font=small, labelfont=bf,
  format=plain, justification=centering,
  skip=6pt,
}
% Per type: \captionsetup[figure]{...}
%           \captionsetup[table]{...}
% Short caption for LOF/LOT:
\caption[Short]{Full long caption text.}
```

CROSS-REFERENCES

```
%-- Placing labels -----
\label{key} % put after captioned item
% Convention: \label{fig:name}, \label{eq:name},
%             \label{tab:name}, \label{sec:name}

%-- Referencing -----
\ref{key} % number: "3" or "4.2"
\pageref{key} % page number: "12"
\eqref{key} % equation: "(3)"
\nameref{key} % section name (hyperref)
\autoref{key} % "Figure 3" (hyperref)

% Always use ~ to prevent bad line breaks:
Figure~\ref{fig:main} % -> "Figure 1"
Section~\ref{sec:intro} % -> "Section 2"
Equation~\eqref{eq:one} % -> "Equation (1)"
Table~\ref{tab:data} % -> "Table 3"
on page~\pageref{fig:main} % -> "page 5"

%-- hyperref: clickable PDF links -----
\usepackage[
  colorlinks = true,
  linkcolor = blue,
  citecolor = green!50!black,
  urlcolor = cyan!70!black,
  pdftitle = {My Document},
  pdfauthor = {My Name},
]{hyperref}

\href{https://example.com}{click here}
\url{https://example.com} % monospace URL
\hyperref[sec:intro]{custom link text}

%-- cleveref: smart automatic naming -----
\usepackage{cleveref}
\cref{fig:main} % -> "fig. 1"
\Cref{fig:main} % -> "Figure 1"
\cref{eq:1,eq:2} % -> "eqs. (1) and (2)"
\crefrange{eq:1}{eq:5} % -> "eqs. (1)-(5)"
```

BIBLIOGRAPHY

BibTeX (traditional)

```
%-- In document body -----
\cite{key}           % -> [1]
\cite[p.~5]{key}     % -> [1, p. 5]
\cite[key1,key2]     % -> [1, 2]
\nocite{key}         % add to list, no marker
\nocite{*}           % add ALL .bib entries

%-- At end of document -----
\bibliographystyle{style}
% Styles: plain alpha abbrv unsrt
%         IEEEtran apalike acm
\bibliography{refs}  % -> reads refs.bib

% Compile order: latex -> bibtex -> latex -> latex

%-- .bib entry types -----
@article{smith2024,
  author = {Smith, J. A. and Doe, J. B.},
  title  = {A Novel Approach},
  journal = {J. Examples},
  year   = {2024},
  volume = {10},
  number = {2},
  pages  = {100--120},
  doi    = {10.1000/xyz123},
}
@book{knuth1984,
  author = {Knuth, D. E.},
  title  = {The {\TeX}book},
  publisher = {Addison-Wesley},
  year   = {1984},
  edition = {1st},
  isbn   = {0-201-13448-9},
}
@inproceedings{conf2024,
  author = {Author, A.},
  title  = {Paper Title},
  booktitle = {Proc. Conf. Name (CONF)},
  year   = {2024},
  pages  = {1--8},
  address = {City, Country},
}
@misc{website,
  author = {Name, F.},
  title  = {Page Title},
  url    = {https://example.com},
  year   = {2024},
  note   = {Accessed: 2024-12-01},
}
% Other types: @phdthesis @mastersthesis
% @techreport @unpublished @manual
```

bibtex (modern alternative)

```
\usepackage[
  backend = biber,      % or bibtex
  style   = numeric,    % ieee, apa, authoryear
  sorting = nyt,        % name, year, title
  maxbibnames = 99,
]{bibtex}
\addbibresource{refs.bib} % in preamble

%-- Cite commands -----
\cite{key}           % [1]
\parencite{key}      % (Smith, 2024)
\textcite{key}       % Smith (2024)
\footcite{key}       % footnote citation
\autocite{key}       % style-dependent

%-- Print bibliography -----
\printbibliography
\printbibliography[type=article,title={Articles}]
\printbibliography[keyword=mykey]

% Compile: pdflatex -> biber -> pdflatex x2
```

CUSTOM COMMANDS & ENVIRONMENTS

```
%-- New command -----
\newcommand{\name}{replacement text}
\newcommand{\name}[n]{use #1, #2, ... #n}
\newcommand{\name}[n][default]{#1 optional, #2 req}
% n = number of arguments (1--9)
% [default] = default for optional 1st argument

%-- Redefine existing -----
\renewcommand{\name}{new definition}
\renewcommand{\arraystretch}{1.5} % row height
\providecommand{\cmd}{def} % no error if exists

%-- Practical examples -----
\newcommand{\R}{\mathbb{R}}
\newcommand{\N}{\mathbb{N}}
\newcommand{\Z}{\mathbb{Z}}
\newcommand{\norm}[1]{\left|#1\right|}
\newcommand{\abs}[1]{\left|#1\right|}
\newcommand{\pd}[2]{\frac{\partial #1}{\partial #2}}
\newcommand{\dd}[2]{\frac{d #1}{d #2}}
\newcommand{\set}[1]{\left\{#1\right\}}
\newcommand{\bra}[1]{\left\langle#1\right|}
\newcommand{\ket}[1]{\left|#1\right\rangle}
\newcommand{\todo}[1]{\color{red}\textbf{[TODO: #1]}}
\newcommand{\highlight}[1]{\colorbox{yellow}{\displaystyle#1}}

%-- Math operators -----
\DeclareMathOperator{\sgn}{sgn}
\DeclareMathOperator{\tr}{tr} % trace
\DeclareMathOperator{\rank}{rank}
\DeclareMathOperator*{\argmin}{arg\,min} % with limits
\DeclareMathOperator*{\argmax}{arg\,max}

%-- New environment -----
\newenvironment{name}[n]{begin code}{end code}

\newenvironment{mybox}
{\begin{center}\begin{tabular}{|c|}\hline}
{\hline\end{tabular}\end{center}}
% Usage: \begin{mybox} content \end{mybox}

%-- Renew environment -----
\renewenvironment{abstract}
{\begin{center}\bfseries Abstract\end{center}}
{\begin{quotation}\small}
{\end{quotation}}

%-- xparse (LaTeX3, more powerful) -----
\usepackage{xparse}
\NewDocumentCommand{\myCmd}{o m}{%
  \IfNoValueTF{#1}{no opt}{#1} and #2}
% Argument types: o=opt[], m=mandatory,
% O{def}=opt with default, s=star
```


CODE LISTINGS

```
%-- Verbatim (no highlighting) -----
\begin{verbatim}
  code as-is, no formatting
\end{verbatim}
\verb|inline code|      \verb+alternative+
\verb*|shows spaces|    % space as _

%-- listings package -----
\usepackage{listings}
\usepackage{xcolor}

\lstset{
  basicstyle      = \ttfamily\small,
  keywordstyle    = \color{blue}\bfseries,
  commentstyle    = \color{gray}\itshape,
  stringstyle     = \color{red!80!black},
  numbers        = left,           % left, right, none
  numberstyle     = \tiny\color{gray},
  stepnumber      = 1,
  numbersep       = 8pt,
  frame           = single,        % single, tb, lines, L
  breaklines      = true,
  tabsize         = 4,
  showspaces      = false,
  showstringspaces = false,
  keepspaces      = true,
  columns         = flexible,
  backgroundcolor = \color{gray!8},
  captionpos      = b,             % b=below, t=above
  morekeywords    = {keyword1},    % add keywords
  deletekeywords  = {\int},        % remove keywords
  escapechar      = @,             % LaTeX inside @
}

% Block listing:
\begin{lstlisting}[language=Python, caption={My Code}]
def fibonacci(n):
    if n <= 1: return n
    return fibonacci(n-1) + fibonacci(n-2)
\end{lstlisting}

% From file:
\lstinputlisting[language=C++]{code.cpp}
\lstinputlisting[language=C++, firstline=5, lastline=20]{code.cpp}

% Inline:
\lstinline[int x = 5;]
\lstinline[language=Python]|print("hi")|

% Define a named style:
\lstdefinestyle{myStyle}{
  language=Java, frame=shadowbox, ...}
\begin{lstlisting}[style=myStyle]

% Supported languages include:
% C C++ Java Python R Matlab Haskell
% HTML XML SQL PHP bash Ruby Fortran
% TeX/LaTeX Perl Scala Swift Go ...
```

BOXES & FRAMES

```
%-- Built-in boxes -----
\fbbox{framed text}
\boxed{a = b}           % math box
\mbox{no-break text}
\makebox[3cm][c]{text}  % fixed-width, centered
\makebox[3cm][l]{text}  % fixed-width, left
\framebox[3cm][r]{text} % framed fixed-width

%-- minipage -----
\begin{minipage}[pos]{width}
  Content here (its own paragraph context)
\end{minipage}
% pos: t=top, c=center, b=bottom

% Side-by-side columns:
\begin{minipage}{0.48\linewidth}
  Left panel
\end{minipage}\hfill
\begin{minipage}{0.48\linewidth}
  Right panel
\end{minipage}

\parbox[t]{5cm}{Multi-line inline box}

%-- tcolorbox -----
\usepackage[most]{tcolorbox}
\begin{tcolorbox}[
  title      = Box Title,
  colback    = blue!5,
  colframe   = blue!50!black,
  fonttitle  = \bfseries,
  arc        = 4pt,
  boxrule    = 0.8pt,
  drop shadow,
]
  Box content here.
\end{tcolorbox}

% Inline colored box:
\tcbbox[colback=red!10,colframe=red]{inline}

%-- mdframed -----
\usepackage{mdframed}
\begin{mdframed}[
  linecolor      = blue, linewidth = 2pt,
  backgroundcolor = gray!10,
  topline=true, rightline=false,
  leftmargin=0pt, innerleftmargin=6pt,
]
  Framed content here.
\end{mdframed}
```

TIKZ BASICS

```
\usepackage{tikz}
\usetikzlibrary{arrows,shapes,positioning,calc}

\begin{tikzpicture}[scale=1, options]
...drawing commands...
\end{tikzpicture}

%-- Lines & Paths -----
\draw (0,0) -- (2,2);           % line
\draw[->] (0,0) -- (3,0);       % arrow ->
\draw[<->] (0,0) -- (0,2);      % <->
\draw[->,thick,red] (0,0) -- (2,1);
\draw (0,0)--(1,1)--(2,0)--cycle; % closed path

%-- Shapes -----
\draw (1,1) circle (1cm);
\draw (0,0) rectangle (3,2);
\draw (1,1) ellipse (2cm and 1cm);
\draw (0,0) arc (0:90:1cm); % arc start:end:radius

%-- Fill & Style -----
\fill[blue!30] (0,0) circle (1cm);
\filldraw[fill=green!20,draw=green] (0,0) rectangle (1,1);
\shade[ball color=blue] (0,0) circle (1cm);

%-- Nodes -----
\node[circle,draw,fill=blue!20] (A) at (0,0) {A};
\node[rectangle,draw,rounded corners] (B) at (3,0) {B};
\draw[->] (A) -- (B);
\draw[->] (A) to[bend left=30] (B); % curved

% Node options:
% draw, fill=color, circle, rectangle, diamond,
% rounded corners, minimum width=, minimum height=,
% text=, font=, align=center, inner sep=2pt

%-- Relative positioning (positioning lib) -
\node[right=1cm of A] (B) {B};
\node[below=0.5cm of B, draw] (C) {C};

%-- Text labels -----
\node at (1,1) [label text];
\node[above=2pt] at (1,1) {Above};
\node[label=above:\small Top] at (0,0) {};

%-- Coordinates -----
\coordinate (P) at (1,1);
\fill (P) circle (2pt); % dot

%-- calc library: arithmetic -----
\coordinate (M) at ($(A)!0.5!(B)$); % midpoint
\coordinate (Q) at ($(A)+(1,0)$); % shifted

%-- Grid -----
\draw[step=1cm,gray!30,thin] (0,0) grid (5,4);

%-- Global styles -----
\tikzset{
  myarrow/.style = {>stealth,thick,blue},
  mynode/.style = {circle,draw,fill=blue!20,
    minimum size=8mm},
}
\draw[myarrow] (0,0) -- (2,0);
\node[mynode] at (1,0) {X};

%-- Flowchart example -----
\tikzset{
  block/.style = {rectangle,draw,fill=blue!10,
    minimum width=2.5cm,minimum height=0.7cm,
    rounded corners=3pt},
  decision/.style = {diamond,draw,fill=green!10,
    minimum width=1.5cm,aspect=2},
}
\node[block] (start) at (0,0) {Start};
\node[decision,below=0.8cm of start] (d) {Cond?};
\node[block,right=1cm of d] (yes) {Yes path};
\node[block,left=1cm of d] (no) {No path};
\draw[->] (start) -- (d);
\draw[->] (d) -- node[above,small]{Y} (yes);
\draw[->] (d) -- node[above,small]{N} (no);
```

MISCELLANEOUS

Counters

```
% Built-in: page section subsection
% figure table equation footnote enumi-iv

\setcounter{section}{3} % set to value 3
\addtocounter{page}{2} % add 2
\stepcounter{equation} % increment by 1
\value{counter} % get numeric value
\thecounter % print formatted
\thesubsection % prints "2.3"

% New counter:
\newcounter{mycount}
\setcounter{mycount}{0}
\stepcounter{mycount}
Count: \themycount.
\refstepcounter{mycount} % allows \label
\label{cnt:this} % now referenceable
```

Lengths

```
% Units: pt mm cm in em ex bp pc sp
% em = width of M in current font
% ex = height of x in current font

% Predefined lengths:
\textwidth \linewidth \columnwidth
\textheight \paperwidth \paperheight
\baselineskip \parindent \parskip
\tabcolsep \arraycolsep \arrayrulewidth
\columnsep \columnseprule

% Modify:
\setlength{\parskip}{6pt}
\setlength{\parindent}{0pt}
\addtolength{\textwidth}{1cm}

% Define new:
\newlength{\mylen}
\setlength{\mylen}{2cm}
\the\mylen % print: "56.9055pt"

% Rubber lengths (stretchable):
\setlength{\parskip}{4pt plus 2pt minus 1pt}
\hspace{\fill} \hspace{\stretch{2}}
```

Footnotes & Marginal Notes

```
Text with note\footnote{Footnote text here.}
\footnote[5]{With specific number.}

% Manual (place separately):
See here\footnotemark
\footnotetext{The text is here.}

% Margin notes:
\marginpar{Note in margin}
\marginpar[raggedleft Left]{Right} % twoside
```

Input / Include

```
\input{filename} % direct paste (no newpage)
\include{filename} % with clearpage, trackable
\includeonly{ch1,ch2} % only compile these files
% \include cannot be nested
```

SI Units (siunitx)

```
\usepackage{siunitx}
\SI{9.81}{\metre\per\second\squared} % 9.81 m/s2
\SI{1.5e8}{\kilo\metre} % 1.5e108 km
\si{joule\per\mole\per\kelvin} % units only
\num{1234567.89} % formatted number
\ang{45} % 45deg
\ang{45;30;20} % 45deg 30' 20''
% Units: \metre \kilogram \second \ampere \kelvin
% \mole \candela \hertz \newton \joule \watt
% \pascal \volt \ohm \coulomb \tesla \henry
```

Conditional & Document Info

```
\usepackage{ifthen}
\newboolean{draft}
\setboolean{draft}{true}
\ifthenelse{\boolean{draft}}
{Draft text}
{Final text}
\ifthenelse{\value{page}=1}{...}{...}

% Useful document commands:
\today           % "January 1, 2025"
\LaTeX{}         % LaTeX logo
\LaTeXe{}        % LaTeX2e logo
\TeX{}           % TeX logo
\thepage         % current page number
\thesection      % current section number
\theequation     % current equation number
\jobname         % source file name (no extension)
```

Phantom / Strut / Rules

```
% Invisible space boxes:
\phantom{text}    % exact space of "text"
\hphantom{text}   % horizontal space only
\vphantom{X}      % vertical height only (aligning)

% Visible rules:
\rule{3cm}{0.4pt} % horizontal rule: {width}{height}
\rule{0pt}{12pt}  % invisible strut (vertical space)
\strut            % full line height invisible box

% Force break or keep together:
\linebreak        \nolinebreak \allowbreak
\mbox{keep together} % no breaks inside
```

Troubleshooting Guide

```
%-- Common Errors -----
% "Undefined control sequence"
%   -> typo in \commandname
% "Missing $ inserted"
%   -> math symbol used outside math mode
% "Runaway argument"
%   -> missing closing brace }
% "Overfull \hbox (Xpt too wide)"
%   -> line too wide for column
% "Too many }'s" or "Extra }"
%   -> unmatched braces
% "File X not found"
%   -> missing package or wrong path

%-- Fixes -----
\sloppy           % relax line-breaking strictness
\usepackage{microtype} % better typography
\hyphenation{cal-cu-la-tion} % manual hyphenation
\linebreak        % force line break
\nolinebreak      % prevent line break
\pagebreak        % force page break
\nopagebreak      % prevent page break

%-- Debug helpers -----
\listfiles        % list all loaded packages
% Run pdflatex twice for TOC/refs to update
% Use latexmk -pdf for fully automatic passes
```

Quick Math Patterns

```
% Newton-Raphson:
x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}

% Taylor series:
f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n

% Euler's identity:
e^{i\pi} + 1 = 0

% Binomial theorem:
(x+y)^n = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k

% Fourier transform:
\hat{f}(\xi) = \int_{-\infty}^{\infty} f(x) e^{-2\pi i x \xi} dx

% Gradient, divergence, curl:
\nabla f = \left( \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)
\nabla \cdot \vec{F} \quad \quad \nabla \times \vec{F}

% Aligned system (use & to align = signs):
\begin{align*}
(x+y)^2 &= x^2 + 2xy + y^2 & \\
(x-y)^2 &= x^2 - 2xy + y^2 & \\
(x+y)(x-y) &= x^2 - y^2 &
\end{align*}
```